INCH-POUND

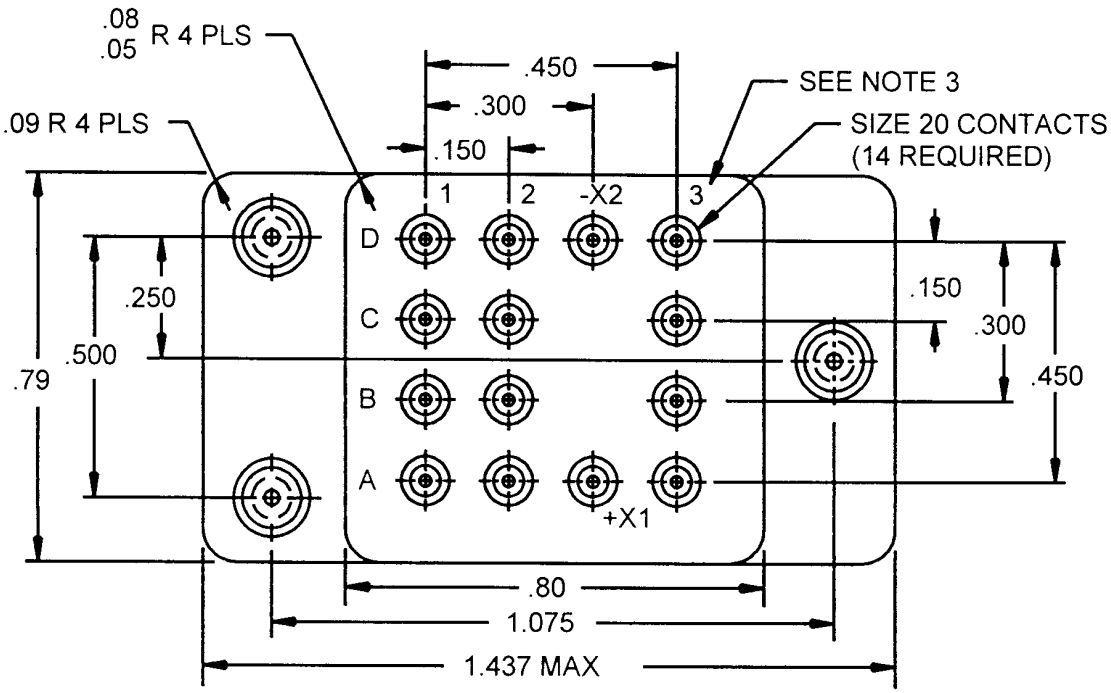
MIL-DTL-12883/44D
18 February 2003
SUPERSEDING
MIL-PRF-12883/44C
28 June 1996

DETAIL SPECIFICATION SHEET

SOCKETS, PLUG-IN ELECTRONIC COMPONENTS, SOCKET FOR RELAYS, 4-POLE, 5 AMPERES
(MIL-PRF-6106 AND MIL-PRF 83536)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein
shall consist of this specification and MIL-DTL-12883.

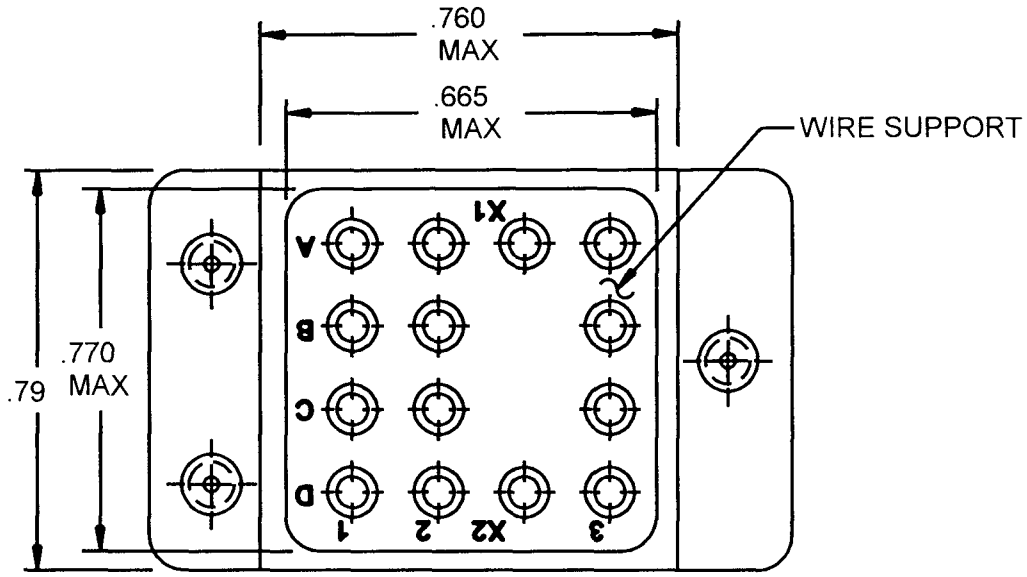


MATING FACE

FIGURE 1. Socket configurations.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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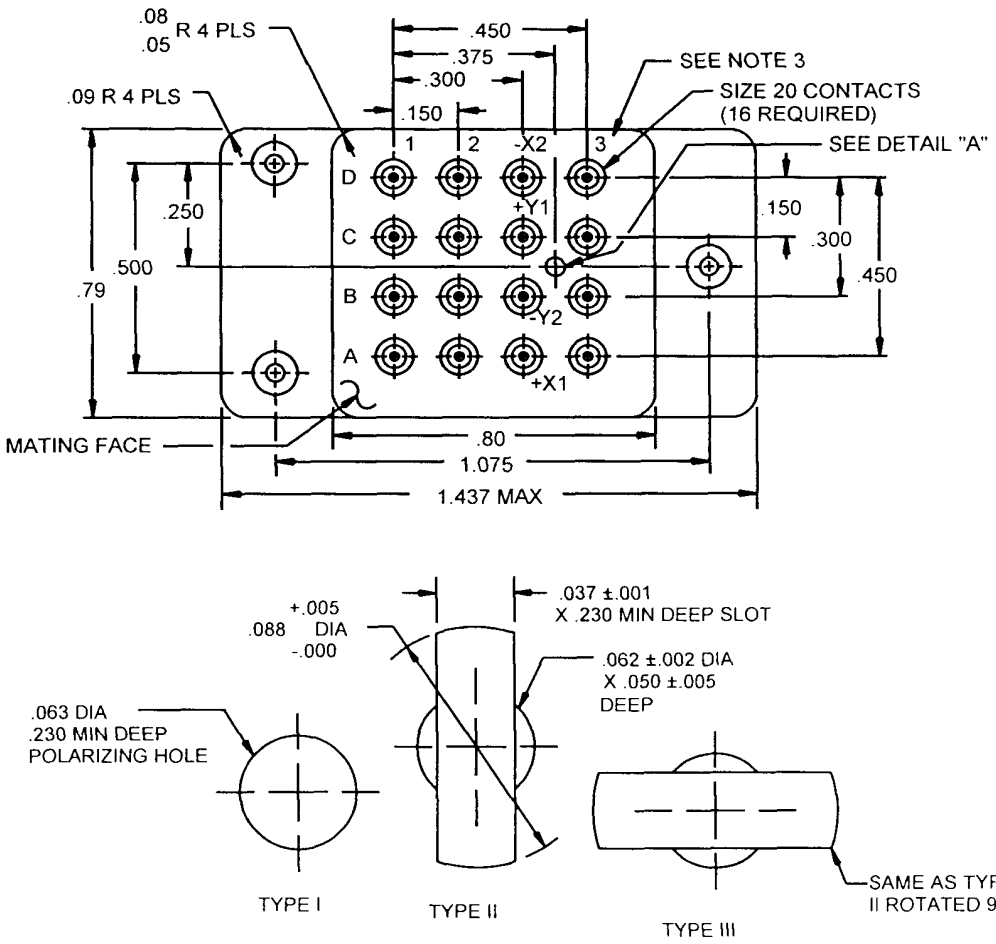


BOTTOM VIEW

(-01)

FIGURE 1. Socket configurations – Continued.

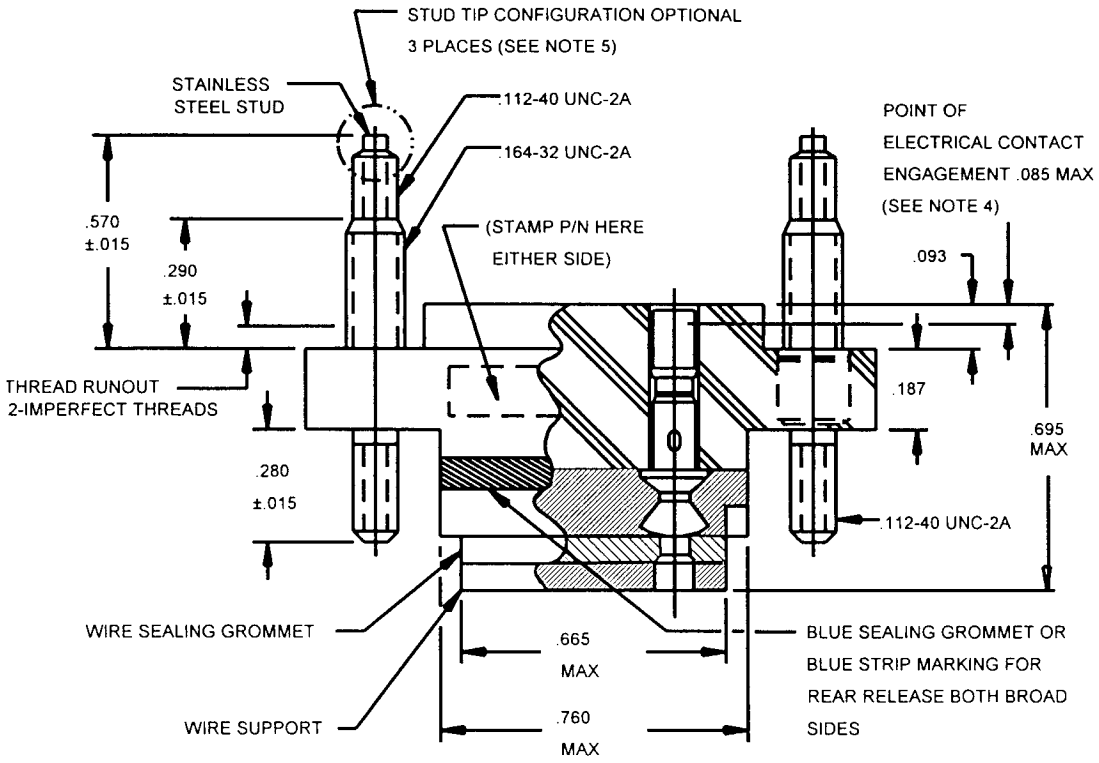
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(-02, -03, and -04)

FIGURE 1. Socket configurations – Continued.

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(01, -02, -03, and-04)

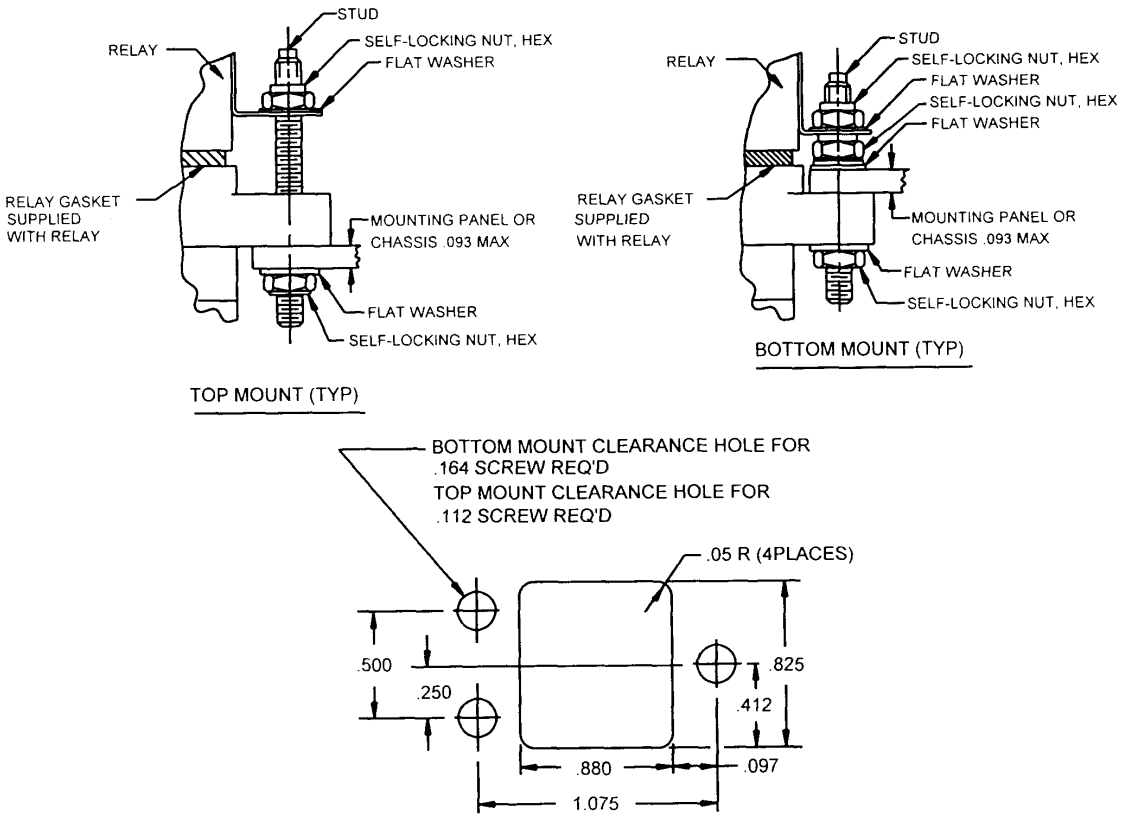
Inches	mm	Inches	mm	Inches	mm
.001	0.03	.088	2.24	.375	9.53
.002	0.05	.093	2.36	.450	11.43
.005	0.13	.112	2.84	.500	12.70
.015	0.38	.150	3.81	.570	14.48
.037	0.94	.164	4.17	.665	16.89
.05	1.3	.187	4.75	.695	17.65
.050	1.27	.230	5.84	.760	19.30
.062	1.57	.250	6.35	.770	19.59
.063	1.60	.280	7.11	.79	20.07
.08	2.0	.290	7.37	1.075	27.31
		.300	7.62	1.437	36.50

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .005 inch (0.13 mm) for three place decimals and ± .01 inch (0.25 mm) for two place decimals.
3. Marking shall be characters, which are molded .035 inch (0.89 mm) minimum. Ink marking optional (see MIL-STD-1285).
4. Point of electrical contact engagement, from mating face of socket insulator to the socket contact.
5. The configuration of the stud tip shall be optional provided it meets all requirements of this specification.
6. Metric equivalents are given for general information only.

FIGURE 1. Socket configuration – Continued

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Inches	mm
.05	1.3
.093	2.36
.097	2.46
.112	2.84
.164	4.17
.250	6.35
.412	10.46
.500	12.70
.825	20.96
.880	22.35
1.075	27.31

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ± .005 (0.13 mm) for three place decimals and ± .01 (0.25 mm) for two place decimals.

FIGURE 2. Mounting (hardware and panel cutout).

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REQUIREMENTS:

Dimensions and configurations: See figures 1, 2, 3, and table I.

Insulator: Diallyl phthalate, in accordance with ASTM D5948, type SDG-F, or any glass filled thermoplastic material in accordance with ASTM D5204.

Color: Material color shall be optional providing that the color provides a contrasting background for the blue sealing grommet or blue color bands indicating rear release contacts.

Grommet: Silicon rubber.

Mounting hardware: Corrosion resistant steel or steel with cadmium/chromate finish.

Electrical

Test gauge: A mating relay in accordance with table I shall be used as a test gauge or mating connector.

Insulation resistance: 1,000 megohms minimum.

Dielectric withstanding voltage:

Sea level: Test voltage: 1,000 V rms.

High altitude, 80,000 feet (24,384 meters): Test voltage: 500 V rms.

Contact resistance: In accordance with MIL-C-39029/101 (see table I).

TABLE I. Dash numbers and configurations.

Dash number/socket configuration	Contact size mating end wire barrel		Number of contacts	Contact designation M39029/101	Mating relay MIL-PRF-83536 1/	Polarization (see figure 1 detail A)
-01	20	20	14	-553	/5-019, -022, -014, /6-003, -006, -011, -014, 019, -022, -029, -032	None
-02	20	20	16	-553	/7-022	I
-03	20	20	16	-553	TBA	II
-04	20	20	16	-553	TBA	III

1/ Reference MIL-PRF-83536 for supersession data on MIL-PRF-6106 relays.

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Environmental:

Temperature range:

Operating temperature range -70ºC to ± 125 ºC.

Wire sealing: A resilient socket grommet is permanently bonded to the wire entry face of the socket so as to provide moisture sealing capabilities, contact M39029/101-553 wire sealing range diameter .040 min to .083 max, (1.02 to 2.11 mm), for AWG size 20 wire (for coil contacts) and AWG size 20 wire (for load contacts).

Contacts: Contacts shall be in accordance with table I. Contacts shall be crimp removable type rear release and accept relay pins size 20 (.039 through .041 inch diameter (0.99 through 1.04 mm)).

Insertion and withdrawal force: The insertion and withdrawal forces shall be as specified in table II.

TABLE III. Insertion and withdrawal forces.

Condition		M12883/44	
		-01	-02 through -04
Initial	Insertion force (max)	16 lbf (71.2 newton)	19 lbf (84.5)
	Withdrawal force (min)	.61 lbf (2.7 newton)	.70 lbf (3.1 newton)
After 10 insertions and withdrawals, before vibration	Insertion force (max)	18 lbf (80.0 newton)	22 lbf (97.7 newton)
	Withdrawal force (min)	.52 lbf (2.3 newton)	.60 lbf (2.7 newton)
After vibration	Insertion force (max)	18 lbf (80.1 newton)	22 lbf (97.9)
	Withdrawal force (min)	.52 lbf (2.3 newton)	.60 lbf (2.7 newton)

Durability: Shall be in accordance with MIL-PRF-12883.

Vibration (sinusoidal): In accordance with MIL-STD-202, method 204, test condition G.

a. Except that the frequency range shall be varied logarithmically between the limits of 10 and 3,000 Hz.
b. Except that the procedure of method of 201 of MIL-STD-202, or an equivalent industry standard, may be applied during 10-55 Hz band of the vibration frequency range.

Vibration (random): In accordance with MIL-STD-1344, method 2005, test condition V, letter G, with a duration of 15 minutes. The mating relay shall be used as the test gage.

Shock (mechanical): In accordance with MIL-STD-202, method 213, condition C, except peak value shall be 200 g's.

Torque: Relay sockets and hardware shall be subjected to testing torque as specified in table III. Sockets shall be installed in mounting panel when test torque is applied. No physical damage will be permitted. Torque shall be maintained for a reasonable period of time to insure stud, socket, and associated hardware have not been damaged.

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TABLE III. Torque requirements (installed in panel conditions).

	Torque			
Thread size	Testing		Installation	
	Inch-pounds	Newton-meter	Inch-pounds	Newton-meter
.112-40	8 + 1 -0	0.90 + .11 -0	4 ±1	0.45 ± .11
.164-32	20 +1 -0	2.26 + .11 -0	15 ±1	1.70 ± .11

Socket, contact tools shall be in accordance with table IV.

TABLE IV. Contact tools.

Nomenclature	Part or Identifying Number (PIN)
Crimp tool	M22520/7-01
Positioner	M22520/7-12
Insertion and removal tool	M81969/16-01

Weight: The socket, hardware, and electrical contacts shall have a maximum weight of .055 pounds (25 grams).

Mounting hardware: The mounting hardware shall allow mounting the socket above, or below the panel or chassis (see figure 2), and shall allow mounting and securing the relay to the socket without disturbing the mounted socket or access to the wiring side of the socket. The hardware shall provide the nominal .197 inch (5.00 mm) spacing between socket surface and relay mounting flange, regardless or mounting configuration.

Mounting hardware shall be supplied with the relay socket and shall consist of the following:

Self locking nuts .112-40 (6 each) (.206 max dia x .176 max height).

Flat washers .112 (6 each) (.206 max O.D. x .021 max thickness).

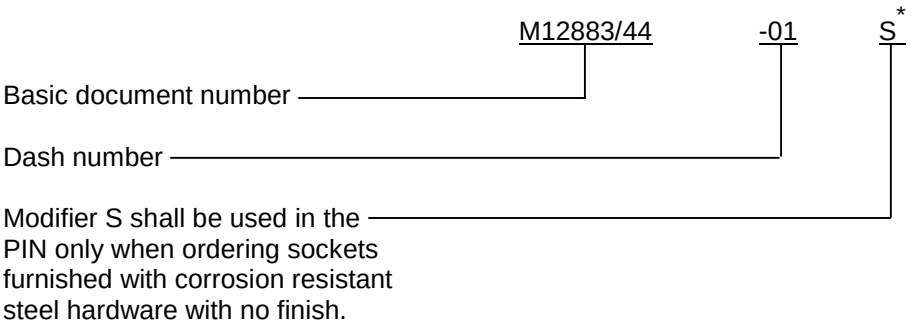
Self locking nuts .164-32 (3 each) (.290 max dia x .190 max height).

Flat washers .164 (3 each) (.290 max O.D. x .019 max thickness).

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PIN: The PIN shall be marked on the socket body as shown in the example (see figure 1). The PIN shall consist of the basic number of this specification sheet and the dash number from table I.

Example:



* For future acquisition of these sockets as of the effective date of revision B, 20 December 1989, parts identified with an “S” modifier shall be corrosion resisting steel (CRS), and parts without an “S” modifier shall be cadmium chromate finish. No mixing of hardware types shall be permitted.

Ordering data: Sockets without contacts may be ordered when so indicated in the ordering data (see MIL-DTL-12883). This applies only to original equipment manufacturers (OEMs) and subcontractors. All direct shipments to the government shall include all applicable contacts and mounting hardware. The PIN to be marked on the socket shall be as shown in the PIN example (see figure 1 and table I).

Revision letters are not used to denote changes due to the extensiveness of the changes.

Active Government PIN	Superseded manufacturer PIN	
	CAGE 58982	CAGE 99699
M12883/44-01	RSE120151	SE405-1020
M12883/44-02	RSE120153	SEL405-1021
M12883/44-03	RSE120155	SEL405-1022
M12883/44-04	RSE120157	SEL405-1023

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy - EC
Air Force – 11
DLA – CC

Review activities:
Army - AR, AT, AV, CR4
Navy - AS, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - CC

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